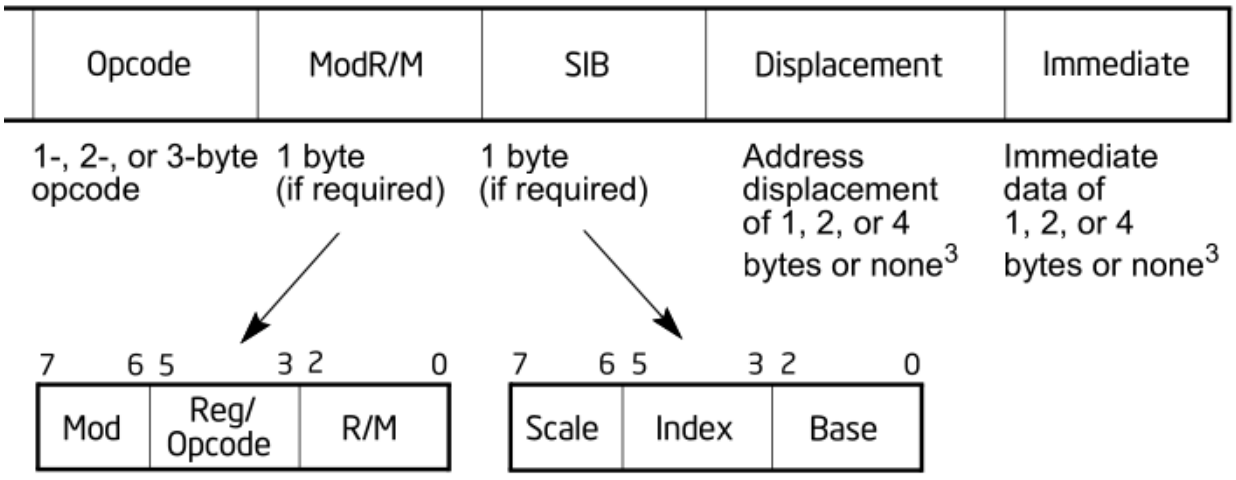
Group Assignment:

Group Members:

1. Sarthak Haribhau Gade (25111015)
2. Sandesh Lalaso Mahanwar (25111035)
3. Jayesh Hiralal Patil (25111026)

Instruction Byte Diagram:



-- **OPCODE:** A primary opcode can be 1, 2, or 3 bytes in length.

-- **MOD R/M:** Takes total 1 byte.

The Mod r/m byte is:

Mod	REG/ OPCODE	R/M
2 bits	3 bits	3 bits

– **SIB**: Takes total 1 byte

The SIB byte will be:

Scale	Index	Base
2 bits	3 bits	3 bits

-- Displacement: If a displacement is required, it can be 1, 2, or 4 bytes.

-- Immediate: An immediate operand can be 1, 2 or 4 bytes.

Encoding Table :
32 bit addressing forms with mod r/m byte

REG	Register
000	**EAX**
001	**ECX**
010	**EDX**
011	**EBX**
100	**ESP**
101	**EBP**
110	**ESI**
111	**EDI**

The Mod r/m byte is:

Mod	REG/ OPCODE	R/M
2 bits	3 bits	3 bits

MOD	Meaning	Displacement
00	Memory Addressing	None
01	Memory Addressing	disp8
10	Memory Addressing	disp32
11	Register-Direct	None

Mod = 00, [Memory, No displacement]

Effective Address	R/M	000	001	010	011	100	101	110	111
[EAX]	000	00	08	10	18	20	28	30	38
[ECX]	001	01	09	11	19	21	29	31	39
[EDX]	010	02	0A	12	1A	22	2A	32	3A
[EBX]	011	03	0B	13	1B	23	2B	33	3B
SIB	100	04	0C	14	1C	24	2C	34	3C
[disp32]	101	05	0D	15	1D	25	2D	35	3D
[ESI]	110	06	0E	16	1E	26	2E	36	3E
[EDI]	111	07	0F	17	1F	27	2F	37	3F

Mod = 01, [Memory, 8 bit displacement]

Effective Address	R/M	000	001	010	011	100	101	110	111
[EAX+disp8]	000	40	48	50	58	60	68	70	78
[ECX+disp8]	001	41	49	51	59	61	69	71	79
[EDX+disp8]	010	42	4A	52	5A	62	6A	72	7A
[EBX+disp8]	011	43	4B	53	5B	63	6B	73	7B
SIB+disp8	100	44	4C	54	5C	64	6C	74	7C
[EBP+disp8]	101	45	4D	55	5D	65	6D	75	7D
[ESI+disp8]	110	46	4E	56	5E	66	6E	76	7E
[EDI+disp8]	111	47	4F	57	5F	67	6F	77	7F

Mod = 10, [Memory ,32 bit displacement]

Effective Address	R/M	000	001	010	011	100	101	110	111
[EAX+disp32]	000	80	88	90	98	A0	A8	B0	B8
[ECX+disp32]	001	81	89	91	99	A1	A9	B1	B9
[EDX+disp32]	010	82	8A	92	9A	A2	AA	B2	BA
[EBX+disp32]	011	83	8B	93	9B	A3	AB	B3	BB
SIB+disp32	100	84	8C	94	9C	A4	AC	B4	BC
[EBP+disp32]	101	85	8D	95	9D	A5	AD	B5	BD
[ESI+disp32]	110	86	8E	96	9E	A6	AE	B6	BE
[EDI+disp32]	111	87	8F	97	9F	A7	AF	B7	BF

Mod=11,[Memory,Register-direct]

Effective Address	R/M	000	001	010	011	100	101	110	111
EAX	000	C0	C8	D0	D8	E0	E8	F0	F8
ECX	001	C1	C9	D1	D9	E1	E9	F1	F9
EDX	010	C2	CA	D2	DA	E2	EA	F2	FA
EBX	011	C3	CB	D3	DB	E3	EB	F3	FB
ESP	100	C4	CC	D4	DC	E4	EC	F4	FC
EBP	101	C5	CD	D5	DD	E5	ED	F5	FD
ESI	110	C6	CE	D6	DE	E6	EE	F6	FE
EDI	111	C7	CF	D7	DF	E7	EF	F7	FF

932 Bit Addressing with SIB

Scale	Index	Base
2 bits	3 bits	3 bits

Scale	[Index Displacement]
00	x1
01	x2
10	x4
11	x8

SIB Index Table :

Bits	Register
000	**EAX**
001	**ECX**
010	**EDX**
011	**EBX**
100	**No Index**
101	**EBP**
110	**ESI**
111	**EDI**

SIB Base Table :

Bits	Register
000	**EAX**
001	**ECX**
010	**EDX**
011	**EBX**
100	**ESP**
101	***EBP**
110	**ESI**
111	**EDI**

Opcode :

Instruction Name	Opcode	Oprand	Register	Memory	Imm	Mod R/M	Imm/ Offset
MOV	C7	r/m32,imm32	1/0	1/0	1	/0	imm32
MOV	89	r/m32,r32	1	1/0	0	/r	–
MOV	8B	r32,r/m32	1	1/0	0	/r	–
ADD	83	r/m32,imm32	1/0	1/0	1	/0	imm32
ADD	81	r/m32,imm8	1/0	1/0	1	/0	imm8
ADD	01	r/m32,r32	1	1/0	0	/r	–
SUB	83	r/m32,imm8	1/0	1/0	1	/5	imm8
SUB	81	r/m32,imm32	1/0	1/0	1	/5	imm32
SUB	29	r/m32,r32	1	1/0	0	/r	–
DEV	F7	r/m32	0	1/0	0	/6	–
INC	FF	r/m32	0	1/0	0	/0	–
DEC	FF	r/m32	0	1/0	0	/1	–
PUSH	50+rd	r32	1	0	0	–	–
PUSH	FF	r/m32	0	1/0	0	/6	–
POP	58+rd	r32	1	0	0	–	–
POP	8F	r/m32	0	1/0	0	/0	–
JMP	EB	rel8	0	0	0	–	rel8
JMP	E9	rel32	0	0	0	–	rel32
JNE	75	rel8	0	0	0	–	rel8

JNE	**0F85**	**rel32**	**0**	**0**	**0**	**–**	**rel32**
RET	**C3**	**–**	**0**	**0**	**0**	**–**	**–**

(*1 = Present in instruction or required,

*0= Not required in instruction or not present,

*0/1= either present or not (both possibilities)if second combined operand will be used first will be skipped.)

#Assembler Directives :

1. section .text → Selects the code section.
2. section .data →Selects the initialized data section.

#Data Directives :

Directive	Data Size	Encoding
DB	8 bits	Direct byte
DW	16 bits	Little-Endian
DD	32 bits	Little Endian

GitHub Links:

1. Gade Sarthak (25111015)

   https://github.com/Sarthak-Gade

2. Mahanwar Sandesh(25111035)

   https://github.com/sandesh8736

3. Jayesh Hiralal Patil(25111026)
   https://github.com/jayeshpatil2003